ECHO
An Evolutive Vocabulary for Collaborative BPM Discussions

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Abstract. Nowadays, Business Process Management (BPM) is considering new approaches that use collaborative environments to involve all types of business process stakeholders in the improvement of the organization's business functions. Nevertheless, when evolving different types of stakeholders, the language gap existing between them is disregarded. Also, these new approaches are only focusing on the top-down strategy since they only allow for such collaboration to occur at business process modeling environments. In this paper, we propose ECHO as an evolutive vocabulary system that focus on the formalization of informal entities supporting both strategies of top-down and bottom-up. ECHO's main objective is to support the evolutive process of formalization of the new business process entities that emerge within the stakeholders' discussions. The main function of those informal entities called concepts is to provide a common language that acts as a "bridge" over the gap existing between the business process stakeholder's individual languages.

Keywords: Business Process Management, Social Software, Tagging, Vocabulary, Discussions, Bottom-Up, Collaboration, Formalization.

1 Introduction

Business processes are known to involve different people, from different organizational units, with different responsibilities and distinct concerns, to collaboratively execute the organization's business functions. Although, the design process of such business functions usually disregards the empiric knowledge owned by those who really operate the business on a daily basis: the business process users. Hence, to achieve a proper management of business processes, we must acknowledge their cross-cutting concern nature: most of the business process concerns cannot be cleanly decomposed into different organizational roles. Business process users own important know-how that business process modelers fail to acquire and manage during the interviews within business process discovery

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or documentation efforts. Similarly, business process users also fail to create and understand models using formal notations that help describing and managing the complexity behind a sound business process orchestration and automation.

Aware of such problematic, new emergent BPM tools are focusing on the involvement of all business process stakeholders. Generally speaking, new collaborative BPM environments such as SAP's Gravity [3], Lombardi's BluePrint [2] and the most recent ArisAlign [1], focus on the management of business processes at the modeling level by providing means to collaboratively involve all different types of business process stakeholders: users, modelers, architects, analysts, developers, managers, or any other type of business process stakeholder. This involvement of all business process stakeholders is also due to the acknowledgment of the benefits of adopting philosophies such as crowd-sourcing, also known as the Wisdom of the Crowds [13], which made the Wikipedia project so successful as we know it today.

In order to provide a realistic example of the benefits of involving all business process stakeholders, we will study the BPM collaboration on emergent changes within a patient's management business process. Normally, when a patient arrives at the hospital reception desk, the receptionist opens a new medical record where his symptoms are registered for triage. After the triage is done, the patient's medical record gets updated and the treatment process begins accordingly to the information captured during the triage. However, about a year ago, a new strain of influenza, the H1N1, has contributed to some emergent changes within that patient management business process. Whenever a patient arrived at a hospital, specific procedures needed to be taken instead of following the standard procedure described before. The first action to be done was to ask the patient if he was feeling any H1N1 influenza symptoms lately. Depending on the answer, the patient would either follow the normal patient management process, or would be given a surgical mask, registered as an influenza patient and forwarded to isolation instead of the common shared waiting room. All these changes in the execution flow of the patient management process emerged from the medical staff and not from business process experts or analysts, reflecting the benefits of involving the business process users directly in the organization's BPM efforts.

Let us now consider that this patient management process was fully supported by a Business Process Management System. If the medical staff was enabled to contribute to the management of that business process, it could create a discussion to explain to the business experts, architects and developers these emergent changes so that they could be properly and formally implemented in the workflow system. Nevertheless, some problems exist within those collaborative BPM discussion environments.

This paper proposes a solution to tackle the problems that emerge within collaborative BPM discussions between the business process stakeholders. Before we propose that solution, we will identify three main problems that emerge from such collaborative BPM discussions. Only after these problems are properly depicted and explained, we will propose a solution that focus on their mitigation: an evolutive vocabulary system. Further, we will present some work related with the solution proposed. Finally, we will conclude by taking some insights about the system presented in this paper, and revealing some paths of future work that we intend to follow.

2 Problems

New collaborative BPM approaches [1; 2; 3] are recognizing the benefits of following social software principles such as egalitarianism[1], which focus on the involvement of all types of business process stakeholders to collaborate in the improvement of the organization's business processes. A common form that supports this collaboration appears as textual discussions between the business process stakeholders. Although this textual discussion collaboration has its advantages and benefits to a more fruitful BPM, it can also arises some problematic issues that we must account for:

Ambiguity
As people communicate via conversations[2], they express themselves on those discussion environments in a unstructured textual way. Those discussions are commonly expressed on a natural language, thus, they represent a potential risk for ambiguity issues. For example, referring to *"the process executed to handle patients"* may, or may not, be the same as referring to *"the process executed whenever a patient arrives at the reception desk"*. In addition to the ambiguity issues that this style of reference may cause, also its extensive length nature makes hard to quickly identify the business process entities being discussed.

Language Gap
Since new collaborative BPM approaches are involving all business process stakeholders, modelers, developers, managers, executors, analysts, and so many other, to participate in the improvement of the organization's business functions, different languages will coexist within their discussions. Business process modelers are more likely to make reference to business process models described with formal notations such as BPMN, while business process users will have more tendency to create references to business process execution entities like data forms and enabled tasks, or business process developers to source-code and more technical issues. All these languages are necessary as they focus on different yet complementing concerns, however, they are likely to create misunderstandings and barriers to mutual-understanding of the business processes when mixed up within the same environment.

Dynamic Vocabulary
Due to the dynamic nature of the organization's business functions, which results in the unpredictable emergence of new business entities, new and different

[1] Relating to, or believing in the principle that all people are equal and deserve equal rights and opportunities.

[2] By conversation is meant the use of a natural language between two or more participants.

concepts constantly appear within the business process stakeholders' vocabulary. Nevertheless, the business process management system may not allow business process users to create ad-hoc entities and associate them to formal business processes in order to fulfill deviation needs. This limitation creates barriers to organized suggestions of new business process entities that could improve the business processes of the organization.

3 Solution Proposal

In order to achieve a more fruitful management of business processes within the context of collaborative BPM discussions, we need to mitigate the set of problems identified in the previous section. Hence, to enhance the business process stakeholder's collaborative productivity within such discussion environments, we propose an evolutive vocabulary system called ECHO[3].

ECHO embraces social production [5; 14], a social software principle that has been recognized to deliver important contributions to BPM approaches [11], enabling business process stakeholders to create and organize information and knowledge about the organization's business functions. To do so, ECHO allows business process stakeholders to create, manage and use semi-structured entities to empower their textual suggestions about the organization's business processes. In ECHO, such semi-structured entities that can be referenced in the business process stakeholders textual discussions are called *concepts*.

Each *concept* acts as a symbolic representation of a business process construct because it can be used to refer either to an entity that already exists and it is formally defined within its respective business process model, or to a completely new entity that has informally emerged from a business process stakeholder's suggestion within a discussion. To allow business process stakeholders to create and refer to concepts that do not formally exist within the BPMS helps in mitigating the dynamic vocabulary issue depicted on the section before. Also, concepts may represent any business process construct at any level of granularity, meaning that a concept can either represent business processes, sub-processes, activities, tasks, data objects, or any other constructs considered in the Business Process Management System (BPMS) domain.

Each *concept* is defined by a *label* that is semantically described and classified by a set of *tags*. ECHO holds such tagging system component to allow the semantic evolution of the concepts that business process stakeholders simultaneously produce and consume within their textual discussions.

This labeled concepts idea mitigates the ambiguity problem depicted in the previous section as it fosters business process stakeholders to use such entities to unambiguously refer to the business process entities in matter. Also, the set of concepts managed by the ECHO system attempts to be the "bridge that crosses"

[3] In reference to the name of the greek mythology nymph who could only repeat the voice of another as a punishment from deceiving Hera from Zeus love affairs with the other nymphs.

the language gap between all different types of business process stakeholders within the same discussion environment. Tagging does not requires any extra knowledge about any other than the natural language.

Typically, the architecture of a tagging system involves defining the rules and relationships between three main entities: users, resources and tags [12].

In our case, the *users* of the tagging system inherent to the ECHO system are all the business process stakeholders; the *resources* are the semi-structured structures "prosumed"[4] by the business process stakeholders that we called concepts; and finally, similarly to concepts, users can also create *tags* or re-use the existing ones to associate to any existing concept with the objective of describing it.

Within the BPM domain, as the nomenclature is uncertain or is constantly evolving due to its dynamic nature, folksonomies appear to be valuable as they provide some structure [12] to the emergent vocabulary of business process entities.

Considering the example given in the introduction section, which is depicted in Figure 1, at least four concepts[5] should be defined within the ECHO system with the following labels: *PatientManagement*, *RegisterPatient*, *Triage* and *UpdatePatientMedicalRecord*. Associating a *label* to a concept is fundamental to understand the semantic direction[6] of its associated tags. At anytime, the ECHO system allows the association of new labels to a given concept so that its semantic direction can be aligned as needed.

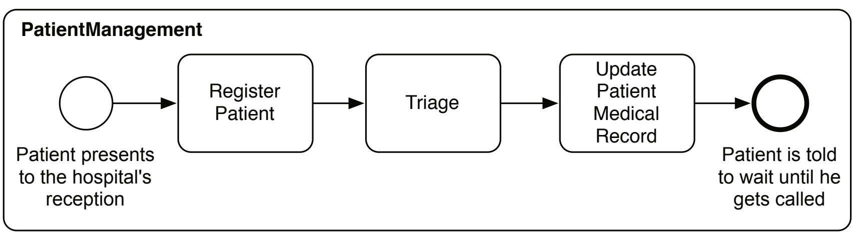

Fig. 1. The patient management business process depicted in this figure is modeled in the Business Process Modeling Notation (BPMN). The business function behind this simple business process allows the initial management of the patient's medical treatment in a hospital. The *PatientManagement* process is composed by three activities: one to register the patient and create his new respective medical process (*Register-Patient*), other to triage the patient in order to obtain a more accurate status of his medical condition (*Triage*), and after these two are executed, another activity that updates the patient's medical record before he gets called to begin the correspondent treatment (*UpdatePatientMedicalRecord*).

[4] Prosumed is a portmanteau that represents the simultaneous action of producing and consuming.

[5] We are omitting the start and ending events for simplicity purposes, however, an analogous approach could be used to define the respective event representative concepts.

[6] A pointer to the meaning the tags are trying to describe.

Now that we have exemplified the *label* construct of a concept and its rationale, we will elaborate on the ECHO's strong folsonomic component mentioned before.

Along with the Web 2.0, a new collaborative approach to help classifying and retrieving content emerged: *folksonomies*. Folksonomies are built within a user-centered approach, where users input keywords and associate them to resources. Such associated keywords describe the content and classify it, so that, in the future, users may easily browse or retrieve the content through those associated tags. We use the `concept` as the entity which wraps both a `label` (e.g. *RegisterPatient*) and the set of `tags` that classify or describe that label (e.g. *register, patient, activity*). However, to provide a more structured classification and description tagging mechanism, the ECHO system supports the idea of having different families of tags.

Tag Families

The ECHO system supports the idea of tag families to provide some organizing structure to the set of tags that it holds. A tag family is a category that identifies a particular type of tag and allows one to quickly identify a particular property of a concept by knowing which tags of that particular family tag are in fact associated to the concept in matter. The setup and configuration of these tag families is highly customizable and flexible, depending entirely on the requirements and needs of the ECHO system administrators. However, we can identify two different types of tag families: categorizing tag families and descriptive tag families.

Categorizing tag families are mutually-exclusive sets of tags that make possible to override previously associated tags of that same family. The classification of the concept accordingly to a particular category, i.e. a particular tag family, is known by looking at which tag of that categorizing tag family is currently associated. On the other hand, we have the descriptive tag families which are non mutually-exclusive sets of tags that focus on describing the concept within the category that tag family represents. Both types of family tags are allowed to evolve as new tags are created and associated to the family tags configured within the ECHO system.

Nevertheless, to better explain this family tags feature and its motivation, we will identify four different families of tags: the first three, *class*, *abstraction* and *formality* tags, are motivated by categorization, while the *descriptive* tags family focus on description instead.

Class Tags

As a categorizing family, *class tags* are mutually-exclusive tags[7] that define the concept's class according to the used business process notation. Class tags allow business process stakeholders, for example, to differentiate either the concept they are referring to is a business process or a business activity. The flexibility in creating this family of tags fosters the ECHO system's independence from the used business process notation.

[7] By mutually-exclusive tags we mean that only one tag of that particular family of tags is valid at a particular point in time.

Based on the simplified business process example depicted in Figure 1, the respective class tags associated with the business process notation would be: *process* and *activity*. Business process stakeholders may create and classify new concepts as belonging to one of those classes, or when needed, create new tags belonging to this family (e.g. data-object) and associate them to any concept.

Abstraction Tags

Abstraction tags are also defined as mutually-exclusive tags, however, they focus on classifying the abstraction level of the concept. Business process stakeholders can distinguish a concept which exists at the execution level, tagged by the `instance` tag, from those which are placed on the modeling higher level of abstraction, tagged by the `type` tag. Abstraction tags help on mitigating the problems inherent from the differences of abstraction levels, by making an explicit distinction between types and their respective instances. Business process stakeholders may explicitly want to refer to particular occurrence where its context of execution is fundamental to a clear comprehension of his suggestion.

Formality Tags

The last set of mutually-exclusive tags are the formality tags, which focus on the degree of formality of a concept. For instance, when a concept is created by a business process stakeholder, the business process entity it refers to is not formally defined yet, hence, it is associated to the `draft` tag. On the other hand, business process entities that are formalized by business process experts and imbued into the business logic of the application are classified as being formal by associating the respective concept to the `formal` tag instead. This family of tags is very important to differ the concepts which are really implemented and formally deployed within the Business Process Management System from those new concepts which are created during the business process stakeholder's suggestions.

Descriptive Tags

Those tags which give semantic value to the concept itself, not regarding any notation-dependency, abstraction or formality concerns, are known as descriptive tags. Descriptive tags allow business process stakeholders to enrich the concept's semantic value through their association to the concept. Examples of descriptive tags are: `patient`, `influenza`, `H1N1`, `check-up` or any other word which is directly related to the concept in matter.

Although the first three tag families were defined as mutually-exclusive, different tags of the same family may be associated to the same concept, however, only the most recently associated tag of each family is considered to be valid on classification-oriented tag families. This grants flexibility to the concept's semantic value, allowing it to evolve and to transform as necessary.

Another important idea to keep in mind is that the ECHO system does not aim to relate or create semantic value between tags. The ECHO system focus on the social production of concept objects and on their semantic value evolution, which results from the collaborative combination and association of the tags described above. The family tags structure is a feature that allows the administrators of the system to easily identify particular folksonomies that emerge from the free user-tagging mechanism.

Now that we have explained all the ECHO constructs (concept, label and tags), we will use the *PatientManagement* process example, depicted in Figure 1, to illustrate the use of the ECHO system a business process stakeholder's suggestion. Let us imagine that a business process user, a doctor for example, creates a new discussion where he states the following:

> *"The* ***PatientManagement*** *process should now consider a new activity* ***%CheckForFlu%*** *to be first executed in order to decide if the patient should be treated normally by executing the* ***RegisterPatient*** *activity or as a special case, needing to execute a* ***%GivePatientSurgicalMask%*** *activity as soon as possible, and then registering him for a specific H1N1 triage so that he can be sent to quarantine."*

In the exemplified suggestion, the business process user recommends two new informal (differentiated as being a draft by the use of the "%" delimiter) activities: *CheckForFlu* and *GivePatientSurgicalMask*. Such concepts were created during that suggestion, and textually related to other formal existing concepts such as the *PatientManagement* process and the *RegisterPatient* activity. The concepts' textual representations are bold and underlined to illustrate the existence of additional information, such as their respective set of associated tags. Nevertheless, this paper does not focus on how that additional information is displayed or digested, and we have only distinguished the draft concept from the remaining formal concepts to depict the emergence of new concepts within textual discussions. The discussion around this suggestion will allow other business process stakeholders to understand it, question it, comment it, and evolve the new concept's semantic value by either associating new tags or new labels. This evolutive aspect of the ECHO system is possible because all the relationships between concept and tag or label objects are defined with temporal awareness. An example of a possible evolution of the *CheckForFlu* concept is illustrated in Figure 2.

This temporal awareness also allows to maintain the evolution history of a concept, meaning that information about previously associated tags or labels is not lost, even when new mutually-exclusive tags are associated. Take the timeline depicted in Figure 2 as an example: both instants `c` and `d` override the concept's label and formality tag, nonetheless, such information is not deleted and is possible to be recreated. This allows to rebuild, as shown in Figure 3, snapshots of the originally labeled *CheckForFlu* concept. Generating a snapshot of a given concept at its time of reference will help the reader of the suggestion to better understand its context and relevance.

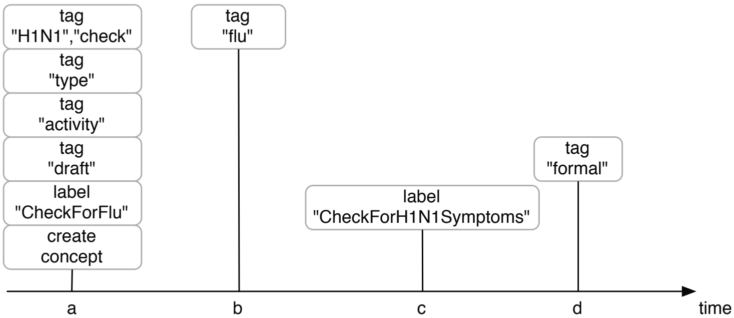

Fig. 2. This figure illustrates a possible evolution of the *CheckForFlu* concept along its lifetime. First, in instant `a`, the *CheckForFlu* concept is created accordingly to the suggestion example. During the concept creation, the business process stakeholder assigns the label "CheckForFlu", tags it as an *activity*, as a *type* and as a new created concept: a *draft*. Also, two descriptive tags are associated in its creation: "H1N1" and "check". Later on instant `b`, another business process stakeholder associates another descriptive tag "flu". Further on instance `c`, another business process stakeholders changes its label to "CheckPatientForH1N1Symptoms". Finally, the instant `d` represents the formalization of the concept, terminating the evolution path between the informal and the formal state of the entity.

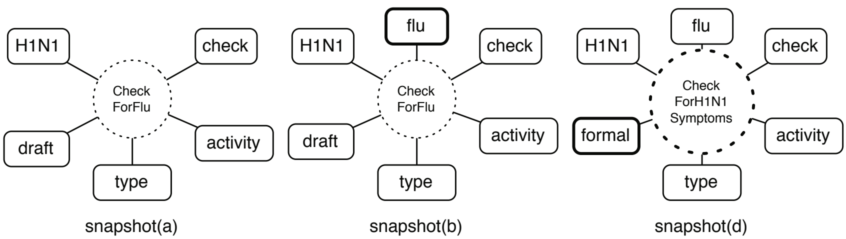

Fig. 3. An important feature of ECHO is that its evolutive aspect allows one to obtain a snapshot of the concept on a given instant in time. Based on the same example depicted in Figure 2, three of the four evolving instants are used to demonstrate this snapshot feature of ECHO.

To conclude, from the ECHO system emerges a weakly-tied[8] type system that business process experts may study, together with the textual suggestions, to better understand the idea behind the business process stakeholders recommendations. Business process experts may look at concepts tagged with the draft tag and understand their semantic value by looking at the concept's evolution and the discussions in which it is referenced. If the discussion and evolution of the patient management business process used in the example had continued, the final business process would have evolved to something like the business process model depicted in Figure 4.

[8] Weakly-tied because there is not a strong conformance between the type-system of the BPMS and the concepts managed by the ECHO system.

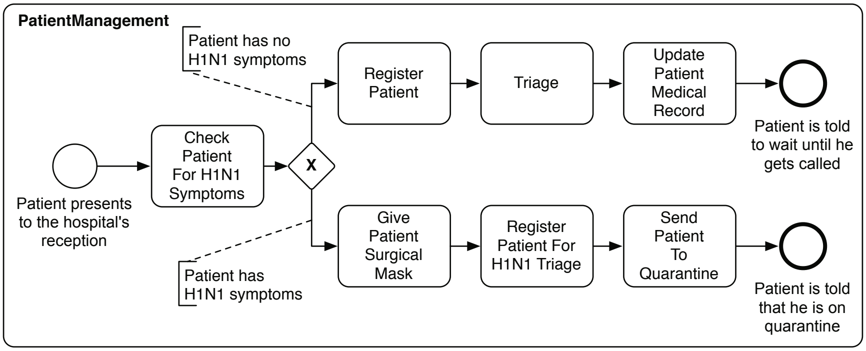

Fig. 4. This figure shows a final version of the *PatientManagement* process after the emergent changes caused by appearance of the H1N1 influenza. According to the concept evolution depicted in Figure 2, a *CheckForH1N1Symptoms* is implemented within the business process, and it becomes the first activity enabled for execution. Depending on the output of that activity, that is, the patient having symptoms of the H1N1 influenza strain or not, the previous flow is taken if there are no symptoms, or in the case there are, a new set of activities will be enabled for execution. Such new activities also appeared in the form of new informal concepts: the *GivePatientSurgeryMask*, *RegisterPatientForH1N1Triage* and *SendPatientToQuarantine* activities.

4 Related Work

In [10], in order to provide some empirical insights and recommendations about activity labeling in process modeling, J. Mendling et al. perceived the usefulness of such labels by analyzing their ambiguity. Also, the existing language gap between business process stakeholders is acknowledged in works like [6] where Cabot et al. proposed a first transformation to bridge the gap from UML/OCL to SBVR specifications.

The family tags were motivated by the work contained in [9] and [7], where two main types of user motivations for tagging are identified: *categorizers* and *describers*. Also, in [8], Jacob identifies the main differences between those two goals when using tagging-systems.

However, there is a main problem with this unrestrictive tagging approach: the semantic of the tags is not explicit as there is no hierarchy and no relationship between the created tags. A method to mitigate this lack of semantics problem is proposed in [15], where users can tag other tags to define the category which they belong to. In the ECHO system, we have used a similar approach, however, instead of tagging other tags, a special "tagable-tag" is used: the concept. Also the family tags serve the purpose to quickly analyze the inherent folksonomies existing within the set of tags associated to a particular concept.

Some interesting metrics are proposed in [4] to identify the type of user motivation. Also, the results shown in [4] reflect that describers are most useful for the emergence of semantics through tagging.

5 Conclusions and Future Work

Business Process Management already considers a large set of approaches and systems that aim for fruitful business process improvements. The use of collaborative environments within those new Business Process Management approaches are a promising attempt to better evolve and improve the business processes of an organization. Focusing on the problems that emerge within the textual discussions between business process stakeholders, this paper proposed a solution based on an evolutive vocabulary system which aims to give support to the formalization process of the informal information that is gathered from those discussions. Based on a strong folsonomic component, the ECHO system allows business process stakeholders to manipulate semi-structured concepts that they may refer to when they want to mention existing business process entities or when they are suggesting new ones. The concepts are allowed to evolve due to the temporal-awareness of the concept's relationships with both label and tag entities. Business process experts may better understand the suggestions made by the business process stakeholders, by analyzing the respective discussions along with the weakly-tied type system that emerges from the set of concepts that the ECHO system holds.

5.1 Future Work

Our future work will focus on the study of another aspects to enhance the ECHO system proposed in this paper.

As ECHO is defined as an independent system to be embedded in a BPM discussion environment, we will study how to provide an Application Programming Interface (API) that will ease the integration of the ECHO system with any Business Process Management System (BPMS).

Another interesting path to investigate is the inference of informal relationships between concepts through the digestion of suggestions. Concepts are related with each other on a unstructured textual form, however, it would be interesting to build a commentary system, on top of the ECHO system, that would concern with those informal relationships between concepts, improving the semantic value of the emergent weakly-tied type system.

Although we are thinking on a well defined API, business process stakeholders are expected to interact with the ECHO system through the means of a Graphical User Interface (GUI). Developers have the freedom to build their own customized GUIs to interact with the ECHO's API, nonetheless, we will study and provide some usability guidelines to improve the User eXperience (UX) of the business business process stakeholders using collaborative Business Process Management Systems (BPMS) that are embedding ECHO.

Finally, when the ECHO system has reached a decent level of maturity, we aim to deploy it on different real Business Process Management Systems to obtain some pragmatic feedback. From such real case studies, we will extract results that will allow us to tweak, tune and improve the system proposed theoretically here in this paper.

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